FINAL PROJECT REPORT

Supply Chain Lead Time Variability Modeling

Develop Predictive Models to Analyze and Forecast Supply Chain Lead Time Variability Using Machine Learning

1. Introduction

Supply chain management plays a crucial role in ensuring that products move efficiently from manufacturers to customers. One of the biggest challenges faced by organizations is the uncertainty in delivery lead times caused by transportation delays, supplier issues, weather conditions, demand fluctuations, and inventory shortages. These uncertainties often result in increased operational costs, delayed deliveries, and poor customer satisfaction.

This project focuses on developing a Supply Chain Lead Time Variability Modeling System using Machine Learning techniques. The system analyzes historical supply chain data to predict future lead times, identify potential bottlenecks, and provide recommendations for improving inventory planning and logistics performance.

The project integrates predictive analytics, time series forecasting, and data visualization to support better business decision-making. By leveraging Artificial Intelligence techniques such as Long Short-Term Memory (LSTM) networks and Gradient Boosting, organizations can forecast delivery delays more accurately and optimize supply chain operations.

The proposed system provides valuable insights for manufacturing, logistics, and retail industries by reducing uncertainty and improving operational efficiency.

2. Problem Statement

Supply chain disruptions have become increasingly common due to unpredictable demand, supplier delays, transportation issues, and global market uncertainties. Traditional forecasting methods often fail to capture complex patterns in historical data, leading to inaccurate delivery predictions and inefficient inventory management.

The objective of this project is to develop a Machine Learning-based predictive analytics platform that:

Predicts future supply chain lead times

Detects delivery bottlenecks

Improves inventory planning

Supports data-driven logistics decisions

Reduces operational costs and delivery risks

3. Objectives

The major objectives of this project are:

Develop a predictive lead time forecasting model.

Perform supply chain data preprocessing and cleaning.

Analyze factors affecting delivery lead time.

Build an LSTM-based forecasting model.

Compare forecasting performance using Gradient Boosting.

Evaluate model performance using MAPE, MAE, and RMSE.

Develop a Supply Chain Analytics Dashboard.

Generate optimization recommendations for inventory planning.

Improve overall supply chain efficiency through predictive analytics.

4. Dataset Description

The dataset contains historical supply chain information collected from logistics and inventory operations. It includes records of orders, suppliers, transportation, inventory levels, and delivery performance.

Dataset Features

Order ID

Product ID

Supplier ID

Order Date

Shipment Date

Delivery Date

Lead Time

Transportation Mode

Inventory Level

Warehouse Location

Order Quantity

Delivery Status

Demand Forecast

Supplier Performance

Delay Status

Dataset Characteristics

Time-series supply chain data

Structured logistics information

Historical delivery records

Inventory-related attributes

Suitable for predictive analytics

Supports machine learning forecasting models

5. Project Architecture

The project follows a modular architecture consisting of the following stages:

Data Collection

Data Preprocessing

Exploratory Data Analysis

Feature Engineering

Time Series Forecasting

Machine Learning Model Development

Model Evaluation

Dashboard Development

Business Recommendations

6. Data Preprocessing

Data preprocessing is a critical step in building an accurate predictive model. Raw supply chain data often contains missing values, duplicate records, inconsistent formats, and outliers that can negatively affect model performance. Therefore, the dataset was cleaned and transformed before training the machine learning models.

Steps Performed

Data Loading

Imported the supply chain dataset into Jupyter Notebook using Pandas.

Data Inspection

Examined dataset structure using .info() and .describe().

Checked data types and summary statistics.

Handling Missing Values

Removed or imputed missing records.

Ensured completeness of important features.

Duplicate Removal

Identified and removed duplicate entries.

Data Formatting

Converted date columns into proper datetime format.

Standardized categorical values.

Feature Scaling

Applied Min-Max Scaling for LSTM model training.

Normalized numerical variables.

Dataset Splitting

Training Dataset: 80%

Testing Dataset: 20%

7. Exploratory Data Analysis (EDA) & Visualization

Exploratory Data Analysis was performed to understand supply chain patterns, identify trends, and discover factors influencing lead time variability.

Visualizations Created

1. Lead Time Distribution

Shows how delivery lead times are distributed across all orders.

Insight: Most deliveries occur within the expected timeframe, while a few delayed shipments create long-tail variability.

2. Monthly Lead Time Trend

Displays average lead time across different months.

Insight: Seasonal fluctuations indicate increased delays during peak demand periods.

3. Supplier Performance Analysis

Compares suppliers based on average delivery lead time.

Insight: Some suppliers consistently deliver on time, while others contribute to delays.

4. Transportation Mode Analysis

Compares lead times for Air, Road, Rail, and Sea transportation.

Insight: Air transport provides the fastest delivery, whereas Sea transport has the highest variability.

5. Inventory Level Analysis

Shows relationship between inventory levels and delivery performance.

Insight: Low inventory levels increase the probability of shipment delays.

6. Actual vs Predicted Lead Time

Compares actual delivery times with model predictions.

Insight: The predicted values closely match actual lead times, demonstrating good forecasting accuracy.

8. Feature Engineering

Feature engineering improves model performance by creating meaningful variables from raw data.

Features Used

Historical Lead Time

Order Quantity

Inventory Level

Supplier Performance Score

Transportation Mode

Warehouse Location

Demand Forecast

Shipping Distance

Delay History

Seasonal Indicator

Encoding Techniques

Label Encoding

One-Hot Encoding

Feature Scaling using Min-Max Scaler

The final processed dataset was optimized for machine learning and deep learning algorithms.

9. Machine Learning Model Development

To improve forecasting accuracy, two advanced machine learning techniques were implemented.

9.1 Long Short-Term Memory (LSTM)

LSTM is a deep learning algorithm specifically designed for sequential and time-series data. It learns long-term dependencies and captures complex temporal patterns in historical lead time records.

Advantages

Handles sequential data efficiently.

Captures long-term dependencies.

High prediction accuracy.

Suitable for forecasting delivery lead times.

9.2 Gradient Boosting Regression

Gradient Boosting is an ensemble machine learning algorithm that combines multiple decision trees to improve prediction accuracy.

Advantages

High predictive performance.

Handles nonlinear relationships.

Reduces prediction errors.

Robust against overfitting when properly tuned.

10. Model Evaluation

The performance of the developed machine learning models was evaluated using standard regression metrics to measure prediction accuracy and reliability.

Evaluation Metrics

Mean Absolute Error (MAE)

Root Mean Squared Error (RMSE)

Mean Absolute Percentage Error (MAPE)

R^2 Score (Coefficient of Determination)

Sample Results

MAPE

15 % out of 14.30 is the accuracy

The LSTM model achieved a MAPE below 15%, meeting the NPCIND evaluation criteria.

The predicted lead times closely matched the actual delivery times.

The model demonstrated strong generalization capability and robustness for forecasting lead time variability.

11. Dashboard Development

A user-friendly Supply Chain Analytics Dashboard was developed using Streamlit to visualize predictions and business insights.

Dashboard Features

Upload supply chain dataset

View historical lead time trends

Predict future lead times

Analyze supplier performance

Monitor inventory levels

Detect logistics bottlenecks

Compare actual vs predicted lead times

Generate optimization recommendations

The dashboard enables decision-makers to monitor supply chain performance in real time and make data-driven operational decisions.

12. Tools and Technologies

Tool

Purpose

Python

Programming Language

Pandas

Data Processing

NumPy

Numerical Computation

TensorFlow

LSTM Model Development

Scikit-learn

Gradient Boosting & Evaluation

Matplotlib

Data Visualization

Plotly

Interactive Charts

Jupyter Notebook

Model Development

Streamlit

Dashboard Development

Git & GitHub

Version Control

13. Challenges Faced

Challenges

Handling missing and inconsistent data

Managing time-series sequences for LSTM

Feature selection for better prediction

Reducing prediction error

Improving model performance

Designing an interactive dashboard

Solutions

Performed thorough data cleaning

Applied feature engineering techniques

Used Min-Max Scaling for LSTM

Tuned hyperparameters for better accuracy

Compared multiple machine learning models

Developed an intuitive Streamlit dashboard

14. Results and Business Impact

The developed system successfully:

Predicted delivery lead times with high accuracy.

Identified major causes of delivery delays.

Improved visibility into supplier performance.

Helped optimize inventory planning.

Reduced uncertainty in logistics operations.

Supported proactive decision-making using predictive analytics.

Business Impact

Reduces operational costs

Improves on-time delivery performance

Enhances customer satisfaction

Optimizes inventory management

Minimizes supply chain risks

Supports strategic planning and forecasting

15. Future Scope

The project can be further enhanced by:

Integrating real-time IoT sensor data

Incorporating weather and traffic information

Using advanced transformer-based forecasting models

Deploying the solution on cloud platforms

Integrating ERP and Warehouse Management Systems

Developing mobile applications for logistics monitoring

Implementing automated inventory optimization

Building Power BI dashboards for enterprise reporting

16. Conclusion

This project demonstrates the practical application of Machine Learning and Deep Learning techniques for predicting supply chain lead time variability. By combining data preprocessing, feature engineering, LSTM networks, and Gradient Boosting algorithms, the proposed system provides accurate lead time forecasts and valuable operational insights.

The developed Supply Chain Analytics Platform enables organizations to identify bottlenecks, optimize inventory levels, and improve logistics efficiency through predictive analytics. The interactive dashboard further enhances decision-making by presenting real-time insights in a user-friendly format.

Overall, this project highlights how Artificial Intelligence can transform traditional supply chain management into a smarter, data-driven, and more efficient system.

17. References

TensorFlow Documentation – https://www.tensorflow.org/

Pandas Documentation – https://pandas.pydata.org/

Scikit-learn Documentation – https://scikit-learn.org/

Streamlit Documentation – https://streamlit.io/

Python Documentation – https://docs.python.org/3/

Research papers on Supply Chain Analytics and Time-Series Forecasting.